Monte Carlo Integration: Recap

Mark M. Fredrickson (mfredric@umich.edu)

Module 1: Integrate — Session 1

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Integrals as expectations

We wish to estimate $\int_a^b g(x)\,dx$. Suppose X is a random variable with:

- Support $[a, b]$ (where either $a \to -\infty$ or $b \to \infty$)
- Density $f(x)$ that is non-zero over $[a, b]$

then

$$\int_a^b g(x)\,dx = \int_a^b \frac{g(x)}{f(x)} f(x)\,dx = \mathrm{E}\left(h(X)\right), \quad h(x) = g(x)/f(x)$$

Sample from X i.i.d. and estimate using $(1/n)\sum_{i=1}^n g(X_i)/f(X_i)$.

Special case: if the target is already $\mathrm{E}\left(g(X)\right)$ for an X we can sample, just sample and average — the density is baked into the sampler (`r` functions; `d` gives the density).

Probabilities are expectations of indicators

$$\mathbf{1}(s) = \begin{cases} 1 & : s \text{ is true} \\ 0 & : s \text{ is false} \end{cases} \qquad \mathsf{E}\left(\mathbf{1}(X \in A)\right) = P(X \in A)$$

In R, indicators are free: `x <= 3`.

The empirical CDF is the sample mean of indicators:

$$\hat{F}_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(X_i \leq t)$$

Why it works

Suppose $X_i \overset{\text{iid}}{\sim} F$, $\mathrm{E}\left(g(X)\right) < \infty$, $\mathrm{E}\left(g(X)^2\right) < \infty$. Then $\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} g(X_i)$ is:

- **Unbiased** for any n: $\mathrm{E}\left(\hat{\theta}_n\right) = \mathrm{E}\left(g(X)\right)$
- **Consistent** (weak law of large numbers): $\hat{\theta}_n \xrightarrow{P} \mathrm{E}\left(g(X)\right)$
- **Asymptotically Normal** (central limit theorem):

$$\frac{\hat{\theta}_n - \theta}{\sigma/\sqrt{n}} \xrightarrow{D} N(0,1)$$

Always report a confidence interval

A Monte Carlo answer is an **estimate**, not an exact solution. By the CLT, an approximate $100(1-\alpha)\%$ interval is

$$\hat{\theta} \pm t_{\alpha/2}(n-1) \times \hat{s}/\sqrt{n}$$

- Continuous $h(X_i)$ values: `t.test(hX)$conf.int`
- Indicator values: `binom.test(sum(hits), n)$conf.int`

The width shrinks like $1/\sqrt{n}$: **each extra digit of precision costs 100 times more samples.**

Error rates and dimension

$$\mathrm{Var}\left(\hat{\theta}_n\right) = \frac{\tau^2}{n}, \quad \tau^2 = \mathrm{Var}\left(g(X)\right) \qquad \Rightarrow \qquad \text{error} = \tau\, n^{-1/2} = O(n^{-1/2})$$

- n sets the **rate**; τ sets the **constant**.
- Grid methods: $O(n^{-2/k})$ in k dimensions — ties Monte Carlo at $k = 4$, loses beyond.
- Monte Carlo's rate **does not depend on the dimension**: sample a random vector, apply g, average.